ISRAEL ALABI

Mechanical Engineering | Controls • Robotics • Mechatronics
City, State | Email | Phone | LinkedIn | GitHub | Portfolio

EDUCATION

McNeese State University — Lake Charle, LA
Bachelor of Science in Mechanical Engineering — Expected May 2027
Minor: Electrical Engineering
GPA: 3.59 / 4.00
Relevant Coursework: Control Systems, System Dynamics, Vibrations, Signals and Systems, Circuits, Electronics, Dynamics.

RESEARCH EXPERIENCE

Research Assistant — August 2026 – Present
Biotechnology lab, McNeese State University
Advisor: Dr. Ahmet Manisali

- Contributing to the development of a smart photobioreactor for cultivation of *Nannochloropsis oculata*, with the long-term goal of integrating data-driven monitoring and control.
- Working with sensing and data collection for key reactor variables including pH, temperature, light intensity, optical density, dissolved oxygen, gas flow rate, liquid level, and salinity.
- Investigating machine-learning and control approaches for algal growth prediction, automated monitoring, and optimization of reactor operating conditions.
- Reviewing research on artificial neural networks, model predictive control, reinforcement learning, and other data-driven methods applied to photobioreactor monitoring and control.
- Contributing to the development of a research manuscript on AI-enabled monitoring and control of photobioreactor systems.

TECHNICAL PROJECTS

Self-Balancing Robot — 2026
Arduino | MPU6050 | PID Control | MATLAB

- Designed and implemented a two-wheeled self-balancing robot using real-time orientation feedback.
- Used MPU6050 orientation measurements as feedback for a controller implemented on an Arduino.
- Generated PWM and direction commands through a dual-channel motor driver to stabilize the robot.
- Experimentally tuned the controller and analyzed the balancing response in MATLAB.

DC Motor Position & Velocity Control — 2026
Arduino | MATLAB | Encoders | Feedback Control

- Implemented closed-loop velocity and position control of a DC motor.
- Calculated angular position and velocity from encoder measurements.
- Experimentally tuned controllers for speed and angular-position tracking.
- Characterized the open-loop relationship between PWM input and steady-state motor velocity.

Gear Design Series — 2026
SolidWorks | Machine Design | Power Transmission

- Designed spur, helical, bevel, worm, and planetary gear systems.
- Constructed involute tooth profiles using calculated gear geometry.
- Applied interference and undercutting criteria when selecting tooth counts.
- Created CAD assemblies to evaluate gear meshing and transmission geometry.

PUBLICATIONS & MANUSCRIPTS

None.

TECHNICAL SKILLS

Programming: C/C++, MATLAB, Python, Html, CSS
Control & Modeling: PID Control, Feedback Control, State-Space Methods, System Dynamics, Simulink
CAD / Mechanical: SolidWorks, Mechanical Design, FEA, Gear Design, Ansys
Electronics / Embedded: Arduino, Microcontrollers, Sensors, Motor Drivers, Oscilloscope, LTspice, Multimeter, Function generator
Fabrication: TIG Welding, Tube Notching, Cutting, Bending, Mechanical Assembly
Software: Git, GitHub, Microsoft Office, Microsoft Excel, Microsoft PowerPoint

ENGINEERING EXPERIENCE

Baja SAE Project — CAD Lead & Drivetrain Team Member Aug. 2024 – Dec. 2025
McNeese Automotive Engineering Club Lake Charles, LA

- Led and supported CAD development for a 4WD Baja SAE all-terrain vehicle designed and built from the ground up.
- Designed propeller shafts, CV axles, engine and transmission mounts, and trailing arm mounting plates in SolidWorks while coordinating fitment with chassis and suspension subsystems.
- Co-developed the pedal system, considering driver ergonomics, packaging constraints, and force transfer within the braking system.
- Designed safety-harness and seat-mounting tabs in accordance with Baja SAE design requirements.
- Participated in suspension-component fabrication using cutting, tube bending, notching, and TIG welding of tubular steel.
- Collaborated with chassis and suspension teams during assembly and design reviews, implementing geometry and mounting revisions to improve subsystem integration and manufacturability.

Dual-Axis Solar Tracker — Senior Design Team Project Jan. 2026 – May 2026
McNeese State University Lake Charles, LA

- Performed structural analysis and wind-load calculations for a dual-axis solar tracking system using engineering mechanics and factor-of-safety methods.
- Analyzed frame members, actuator mounting locations, shafts, brackets, and other structural components under expected loading conditions.
- Conducted bolt sizing and combined loading calculations to evaluate shear, axial loading, and structural safety.
- Used SolidWorks simulation and engineering calculations to support structural design validation and component selection.
- Contributed to the development of the solar-tracking control concept using Arduino-based sensing and actuator-control methods.
- Supported prototype integration and testing as part of the senior design team.

HONORS & AWARDS

None

LEADERSHIP & ACTIVITIES

American Society of Mechanical Engineers (ASME) — Member Jan. 2023 – Present

National Society of Black Engineers (NSBE), McNeese Chapter — Member Jan. 2023 – Present

ADDITIONAL INFORMATION

Languages: English, Yoruba
Professional Memberships: ASME, IEEE, etc.